



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI AMARAVATHI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

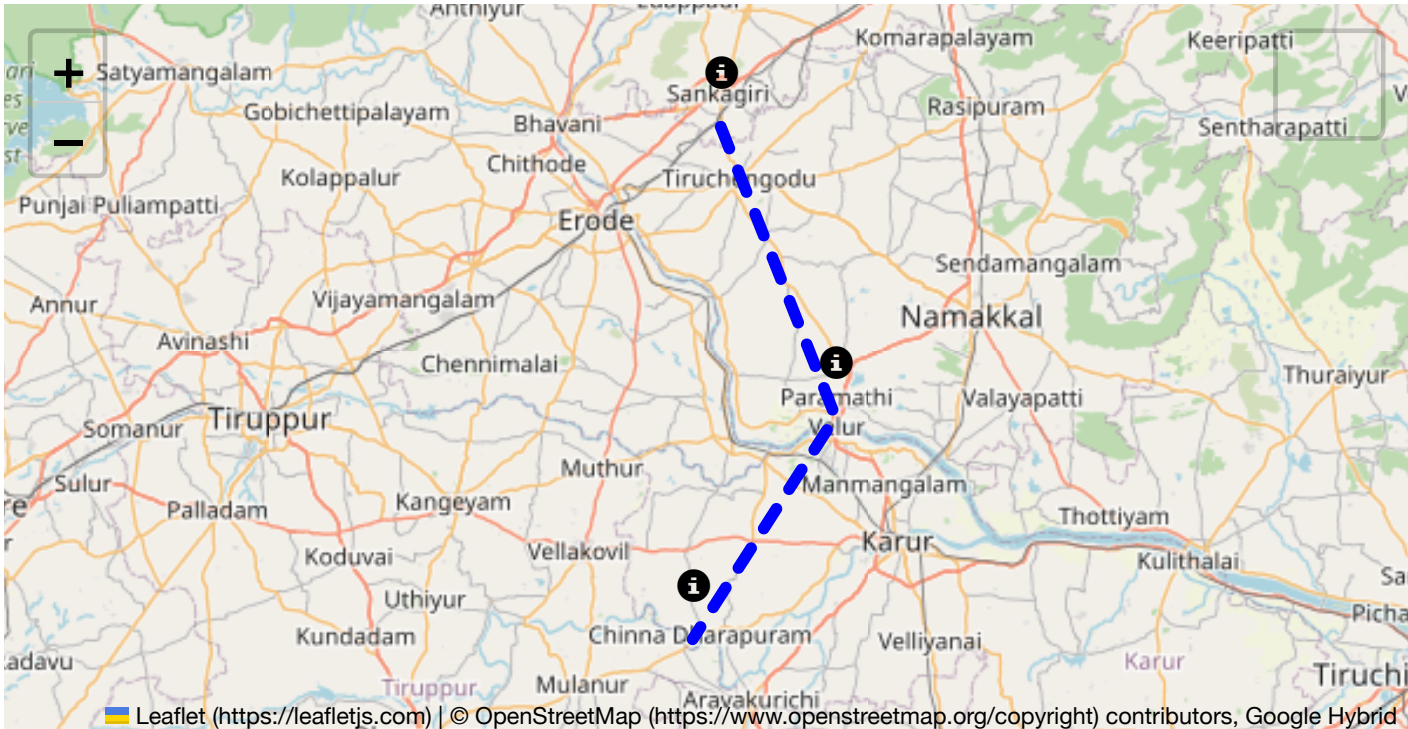
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 95.00 km
Estimated Duration: 1.9 hours
Adjusted Duration (Heavy Vehicle): 2.4 hours

Start: (11.4381, 77.8734)
End: (10.84731, 77.841057)



Welcome to the Journey Risk Management Study

Route Overview

The route spans approximately 95 kilometers and begins at CVQF+23W, Sangagiri, traveling via the Karur - Namakkal Highway, passing through Velur, and concluding at RRXR+2HH, Kalyana Madapam Road, Kodanthur South. The journey typically takes about 1.91 hours for heavy vehicles carrying hazardous materials.

Weather Conditions

- **Typical Weather:** The region experiences a tropical climate with high temperatures, especially from March to May. The monsoon season occurs from June to September, bringing heavy rainfall.
- **Potential Hazards:** During the monsoon, roads can become slippery and may experience waterlogging, potentially slowing down traffic and reducing visibility. Fog can be a concern in the early morning hours during the winter months.

Traffic Patterns

- **Peak Hours:** Traffic congestion is typically higher during morning hours (7:00 AM to 9:00 AM) and evening hours (5:00 PM to 7:00 PM) around major intersections and towns such as Namakkal and Velur.
- **Congestion-Prone Areas:** Traffic tends to bottleneck at town centers and toll plazas.

Road Quality and Infrastructure

- **Assessment:** The majority of the route on the Karur - Namakkal Highway is well-maintained with major sections being well-paved. However, some secondary roads in rural areas may be narrower and in poorer condition.

- **Key Infrastructure:** Adequate signage, rest stops, and gas stations are available along the highway. The availability decreases once you deviate into smaller towns or rural areas.

Alternative Routes for Emergencies

- **Route 1:** An alternative can involve using the Pallipalayam - Pallipalayam Kottai Road to avoid traffic in Velur, reconnecting with the highway at a later point.
- **Route 2:** Use the Vellalayam-Kovai Main Road to bypass Namakkal if there is severe congestion or road blockage.

Regulations for Hazardous Material Transport

- **Local Regulations:** Vehicles transporting hazardous materials must adhere to national safety standards, including proper labeling and the availability of safety equipment. Routes must be pre-approved in certain jurisdictions with specific time restrictions, particularly in urban regions.

Historical Incidents

- **Overview:** There have been occasional incidents involving heavy vehicles, often due to poor visibility and road conditions during severe weather. However, major accidents involving hazardous materials are rare.
- **Safety Measures:** Installment of speed breakers and regular inspection of vehicle fitness are enforced.

Environmental Considerations

- **Sensitive Areas:** The route passes near agricultural lands and small water bodies, necessitating care in preventing spills and environmental contamination.

Communication Coverage

- **Coverage Analysis:** Major sections of the highway have strong mobile network coverage from major Indian providers. However, some rural and less developed sections may have poor to moderate coverage.

Emergency Response Times

- **Response Times:** Response times are generally under 30 minutes in urban centers along the route. In more remote areas, times could extend to an hour or more due to lesser availability of emergency services.

Overall Risk Assessment Summary

In summary, while the primary highway is in good condition and equipped to handle heavy vehicles, caution is necessary during monsoon weather due to potential waterlogging and reduced visibility. Traffic congestion in towns can cause delays, necessitating the knowledge of alternative routes for emergencies. Communication is mostly reliable but spotty in remote areas. Emergency services respond quickly in urban areas but may be limited in rural zones. Proper adherence to regulations is crucial for the safe transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.96304, 78.06131	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44019, 77.87547	15 KM/Hr
4	Turn	Medium	11.39685, 77.88861	30 KM/Hr
5	Turn	High	11.37862, 77.89488	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16764, 78.01565	30 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr
9	Turn	High	11.14480, 78.02291	15 KM/Hr
10	Turn	Medium	11.10125, 78.00603	30 KM/Hr
11	Turn	Medium	11.10012, 78.00502	30 KM/Hr
12	Turn	Medium	10.95701, 78.01249	30 KM/Hr
13	Turn	Medium	10.94431, 78.00335	30 KM/Hr
14	Turn	Medium	10.94317, 78.00168	30 KM/Hr
15	Turn	Medium	10.93382, 77.99870	30 KM/Hr
16	Turn	Medium	10.92666, 77.99911	30 KM/Hr
17	Turn	Medium	10.90391, 77.97828	30 KM/Hr
18	Turn	Medium	10.89755, 77.96991	30 KM/Hr
19	Turn	High	10.88006, 77.95528	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium
19	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium
20	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium
21	hospital	Krishna Medical Centre Hospital	10.9643009, 78.0541215	30 km/h	Medium
22	clinic	Sivabakkiyan Poly Clinic	10.9642254, 78.0549579	30 km/h	Medium
23	hospital	Meganathan Hospital	10.9658739, 78.052368	30 km/h	Medium
24	hospital	vasu medical	10.9638208, 78.0530404	30 km/h	Medium
26	hospital	Prakash Hospital	10.85157, 77.846321	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
25	school	st. antonys matric hr. secondary school	10.9641269, 78.0417919	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.96304, 78.06131



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37862, 77.89488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.10125, 78.00603



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.10012, 78.00502



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.95701, 78.01249



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.94431, 78.00335



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.94317, 78.00168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.93382, 77.99870



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.92666, 77.99911



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.90391, 77.97828



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.89755, 77.96991



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.88006, 77.95528
